Workshop 2 — Design Artifacts and System Modeling

Kintsugi Studio — Online Booking and Business Management Platform

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Table of Contents

1. System context
2. Actor characterization
3. Business rules
4. CRC cards
5. Class diagram
6. Relational database model
7. Mockups
8. Business process models (BPMN)
9. Architecture diagram
10. Design decisions

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1. System Context

Kintsugi Studio operates from a single site in Bogotá and serves approximately 200–230 clients per month. The platform sits between four groups of users and four external services:

- **Clients** discover services and book appointments from mobile or desktop.
- **Professionals** consult their agenda and service history.
- **Administrators** run the daily operation and financial control.
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System boundary: the platform owns clients, services, professionals, appointments, payments, commissions, cash closings, inventory, and notifications. Google Calendar, WhatsApp, the payment gateway, and the email provider are external systems accessed through adapters.

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2. Actor Characterization

Actor	Type	Goals	Permissions	Devices
Visitor	Human, unauthenticated	Learn about the studio and start a booking.	View public content and availability.	Mobile and desktop browser.
Client	Human, authenticated	Book, reschedule, cancel, and review their history.	Read and update their own contact data; create and manage their own appointments.	Mobile and desktop browser.
Professional	Human, authenticated	See their agenda, service history, and reminders.	Read their own agenda and history; add service notes and photos to their own appointments.	Mobile and desktop browser.
Administrator	Human, authenticated	Run the daily operation and financial control.	Manage all operational data except platform-level settings.	Desktop and tablet.
Google Calendar	External system	Synchronize appointments.	Receive and update calendar events through OAuth 2.0.	API.
WhatsApp Business	External system	Deliver confirmations, reminders, and birthday messages.	Receive template messages and webhook events.	API.
Payment gateway	External system	Process online payments when enabled.	Receive payment requests and send webhook events.	API.

Actor	Type	Goals	Permissions	Devices
Email provider	External system	Deliver transactional email.	Receive send requests and report delivery status.	API.

Role boundaries

- The **administrator** can change prices, schedules, clients, and appointments, but cannot alter platform-level configuration.
- The **professional** can see only their own agenda and the clients they have served.
- The **client** can see only their own data and cannot edit their service history.

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3. Business Rules

ID	Rule
BR-01	An appointment occupies the service duration plus the configured preparation buffer.
BR-02	Two appointments for the same professional cannot overlap in time.
BR-03	An appointment must fall within the professional's working hours and outside schedule blocks.
BR-04	A selected slot is held temporarily during the booking flow and released if the booking is not completed within the configured window.
BR-05	Appointment states are `pending` → `confirmed` → `completed`; `cancelled` and `no_show` are terminal states.
BR-06	Only administrators can change prices or cancel any appointment.
BR-07	Professionals see their own agenda and service history; they cannot see financial information.
BR-08	Only administrators can access financial information.

ID	Rule
BR-09	The professional commission is 55 % of the service total, or 60 % when the professional provides their own implements.
BR-10	Commissions are settled weekly.
BR-11	Referral benefit: 5,000 COP discount per referred client; 20,000 COP when the client reaches three referrals within one month of the last service.
BR-12	Birthday benefit: a reminder is sent one week before, and a 10 % discount is valid during the birthday week.
BR-13	The system must warn about possible duplicate clients using normalized phone, email, or document number.
BR-14	Clients cannot edit their data or service history; only administrators can.
BR-15	Sensitive operations are recorded in an audit log, personal data is retained for one year, and administrators must use two-factor authentication.

Operational parameters

- Expense categories: `service`, `rent`, `providers`, `daily expenses`.
- Monthly sales goals are configured per month.
- The studio works with approximately seven providers, mostly on 30-day credit.
- Inventory records entries, exits, and adjustments.
- Petty cash depends on the day and is recorded in the cash closing.
- There is no minimum stock alert in the MVP.
- Only internal users access the platform; providers do not have a portal.

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4. CRC Cards

CRC-01 — Client

Responsibility	Collaborator
Store identity and contact data.	AppUser
Track appointments and service history.	Appointment, AppointmentItem

Responsibility	Collaborator
Record marketing consent and privacy authorization.	ConsentRecord
Accumulate referrals and birthday benefits.	Referral, GiftCard
Support segmentation by frequency and value.	Appointment

CRC-02 — Professional

Responsibility	Collaborator
Store identity, specialties, and display name.	AppUser
Define working hours and schedule blocks.	WorkingHours, ScheduleBlock
Offer services with custom price or duration.	ProfessionalService, Service
Receive commissions.	Commission
Access own agenda and history.	Appointment

CRC-03 — Service

Responsibility	Collaborator
Store name, description, duration, and price.	ServiceCategory
Be offered by one or more professionals.	ProfessionalService
Be included in appointments.	AppointmentItem

CRC-04 — Appointment

Responsibility	Collaborator
Bind a client, a professional, a date, and a time.	Client, Professional
Enforce state transitions and overlap rules.	AvailabilityPolicy
Contain one or more services.	AppointmentItem
Trigger notifications and calendar synchronization.	Notification, CalendarEvent

CRC-05 — AvailabilityPolicy

Responsibility	Collaborator
Calculate free slots from working hours, blocks, and existing appointments.	WorkingHours, ScheduleBlock, Appointment
Apply duration, buffer, and minimum anticipation.	Service, Professional
Validate a proposed appointment.	Appointment

CRC-06 — AppointmentItem

Responsibility	Collaborator
Store the service, price, and duration agreed for the appointment.	Service
Serve as the base for commission calculation.	Commission

CRC-07 — Payment

Responsibility	Collaborator
Record amount, method, status, and gateway reference.	Appointment
Support cash, card, transfer, and gift card payments.	GiftCard, CashClosing
Feed the daily cash closing.	CashClosing

CRC-08 — GiftCard

Responsibility	Collaborator
Generate a unique code and track the balance.	Client
Apply discounts to a booking.	Payment
Expire according to the configured policy.	—

CRC-09 — Commission

Responsibility	Collaborator
Calculate 55 % or 60 % of the service total.	AppointmentItem, Professional
Group commissions by week.	—
Record settlement status and date.	Professional

Responsibility	Collaborator

CRC-10 — CashClosing

Responsibility	Collaborator
Summarize income by payment method and expenses by category.	Payment, Expense
Record the counted cash and the difference.	Expense
Require a justification when the difference is not zero.	—

CRC-11 — InventoryItem

Responsibility	Collaborator
Store SKU, stock, cost, and sale price.	StockMovement
Track entries, exits, and adjustments.	StockMovement
Link to the provider that supplies it.	Provider, ProviderInvoice

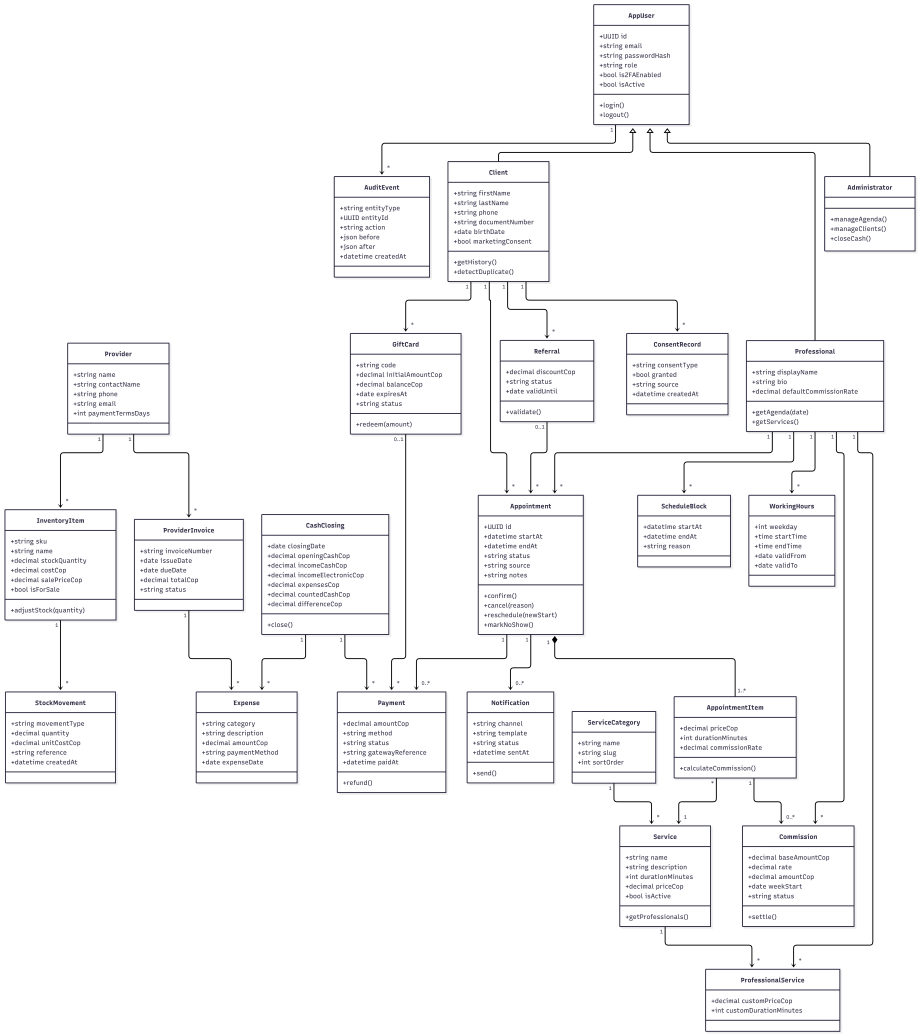
CRC-12 — Provider

Responsibility	Collaborator
Store contact data and payment terms.	ProviderInvoice
Track invoices and accounts payable.	Expense
Supply inventory items.	InventoryItem

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5. Class Diagram

The following class diagram illustrates the system's structure, representing its main classes, attributes, methods, and the relationships between them. It provides an overview of the system's object-oriented design and how its components interact.



Key relationships

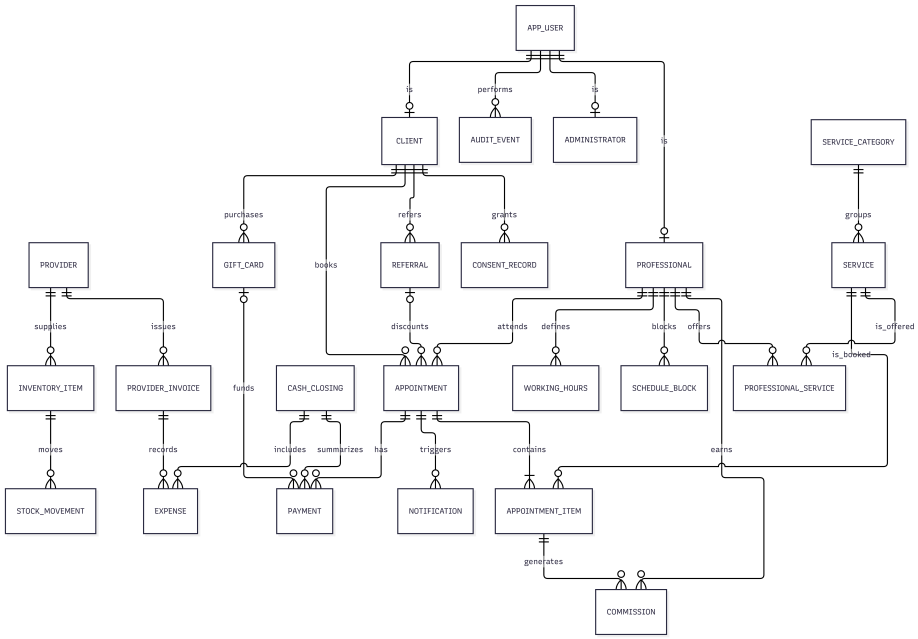
- `AppUser` is specialized into `Client`, `Professional`, `Administrator`.
- An `Appointment` belongs to one `Client` and one `Professional` and contains one or more `AppointmentItem` records.
- `AvailabilityPolicy` is a domain service that collaborates with `WorkingHours`, `ScheduleBlock`, and `Appointment`.
- `Commission` is calculated from `AppointmentItem` and settled per `Professional` per week.
- `CashClosing` aggregates `Payment` and `Expense` records for a date.

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6. Relational Database Model

6.1 Entity-relationship diagram

The following Entity-Relationship diagram represents the system's database structure, identifying its main entities, attributes, and relationships. It provides a conceptual overview of how data is organized, connected, and managed within the database.



6.2 Table definitions

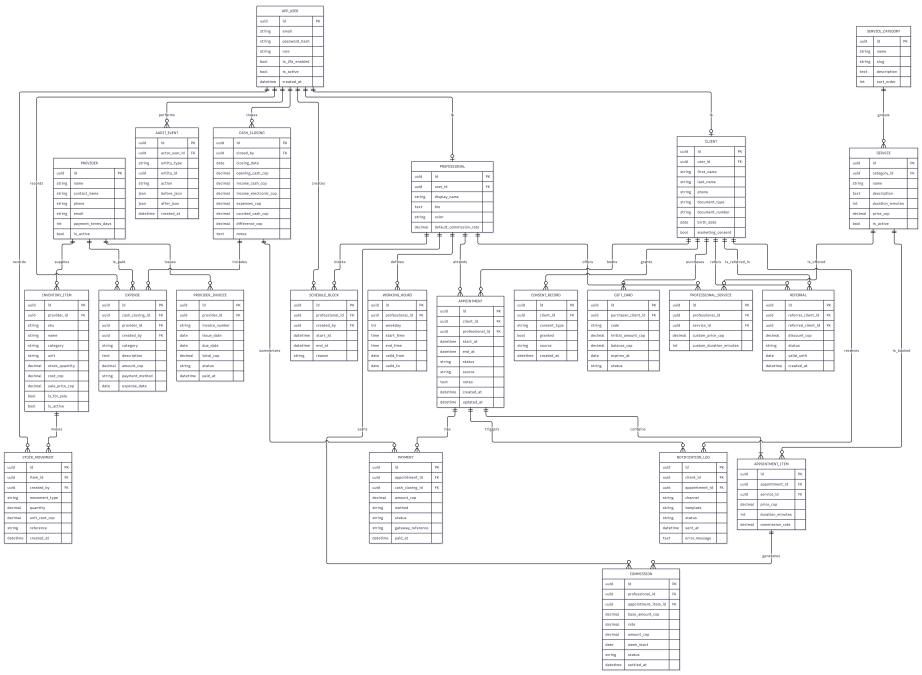


Table	Primary key	Main columns	Foreign keys
`app_user`	`id`	`email`, `password_hash`, `role`, `is_2fa_enabled`, `is_active`, `created_at`	—
`client`	`id`	`first_name`, `last_name`, `phone`, `document_type`, `document_number`, `birth_date`, `marketing_consent`	`user_id` → `app_user`
`professional`	`id`	`display_name`, `bio`, `color`, `default_commission_rate`	`user_id` → `app_user`
`service_category`	`id`	`name`, `slug`, `description`, `sort_order`	—

Table	Primary key	Main columns	Foreign keys
service	id	name, description, duration_minutes, price_cop, is_active	category_id → service_category
professional_service	id	custom_price_cop, custom_duration_minutes	professional_id → professional, service_id → service
working_hours	id	weekday, start_time, end_time, valid_from, valid_to	professional_id → professional
schedule_block	id	start_at, end_at, reason	professional_id → professional, created_by → app_user
appointment	id	start_at, end_at, status, source, notes, created_at, updated_at	client_id → client, professional_id → professional
appointment_item	id	price_cop, duration_minutes, commission_rate	appointment_id → appointment, service_id → service
payment	id	amount_cop, method, status, gateway_reference, paid_at	appointment_id → appointment, cash_closing_id → cash_closing
gift_card	id	code, initial_amount_cop, balance_cop, expires_at, status	purchaser_client_id → client
referral	id	discount_cop, status, valid_until, created_at	referrer_client_id → client, referred_client_id → client

Table	Primary key	Main columns	Foreign keys
commission	id	base_amount_cop, rate, amount_cop, week_start, status, settled_at	professional_id → professional, appointment_item_id → appointment_item
cash_closing	id	closing_date, opening_cash_cop, income_cash_cop, income_electronic_cop, expenses_cop, counted_cash_cop, difference_cop, notes	closed_by → app_user
expense	id	category, description, amount_cop, payment_method, expense_date	cash_closing_id → cash_closing, provider_id → provider, created_by → app_user
inventory_item	id	sku, name, category, unit, stock_quantity, cost_cop, sale_price_cop, is_for_sale, is_active	provider_id → provider
stock_movement	id	movement_type, quantity, unit_cost_cop, reference, created_at	item_id → inventory_item, created_by → app_user
provider	id	name, contact_name, phone, email, payment_terms_days, is_active	—
provider_invoice	id	invoice_number, issue_date, due_date, total_cop, status, paid_at	provider_id → provider

Table	Primary key	Main columns	Foreign keys
notification_log	id	channel, template, status, sent_at, error_message	client_id → client, appointment_id → appointment
audit_event	id	entity_type, entity_id, action, before_json, after_json, created_at	actor_user_id → app_user
consent_record	id	consent_type, granted, source, created_at	client_id → client

Integrity rules

- client.user_id, professional.user_id, and app_user.email are unique.
- professional_service has a unique constraint on (professional_id, service_id).
- appointment uses a database constraint or a transaction-level lock to prevent overlaps.
- payment.gateway_reference is unique when the gateway is used.
- gift_card.code is unique.
- cash_closing.closing_date is unique per day.
- inventory_item.sku is unique.
- Deletes are logical (is_active) for clients, professionals, services, and inventory items.

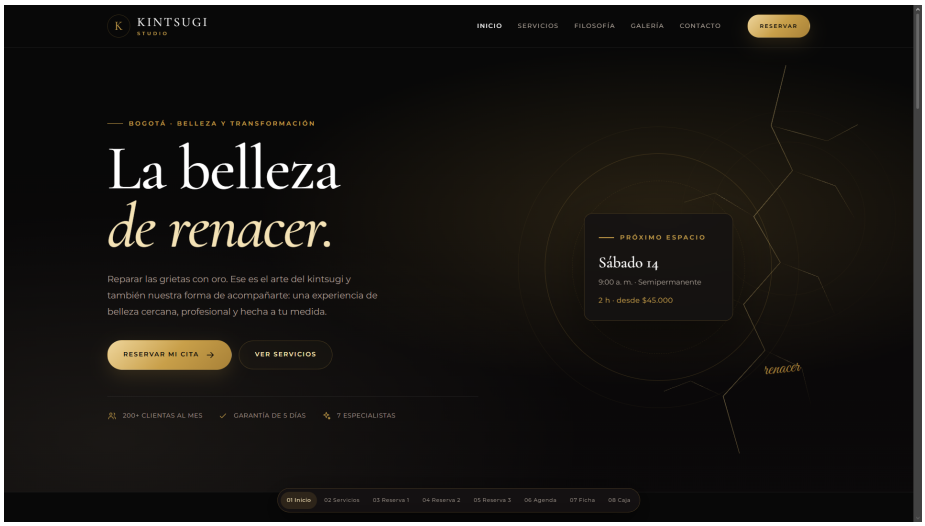
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7. Mockups

The mockups follow the brand manual: deep black #080808, Kintsugi gold #C9A04A, champagne #F4E2B5, pearl white #FFFFFF, and graphite #292526; typography uses Montserrat, Cormorant Garamond, and Great Vibes.

7.1 Home page

Purpose: communicate the brand, the value proposition, and the main booking action. **Elements:** hero image, slogan, primary "Book now" button, featured services, gallery preview, testimonials, contact section, footer with social media and policies.



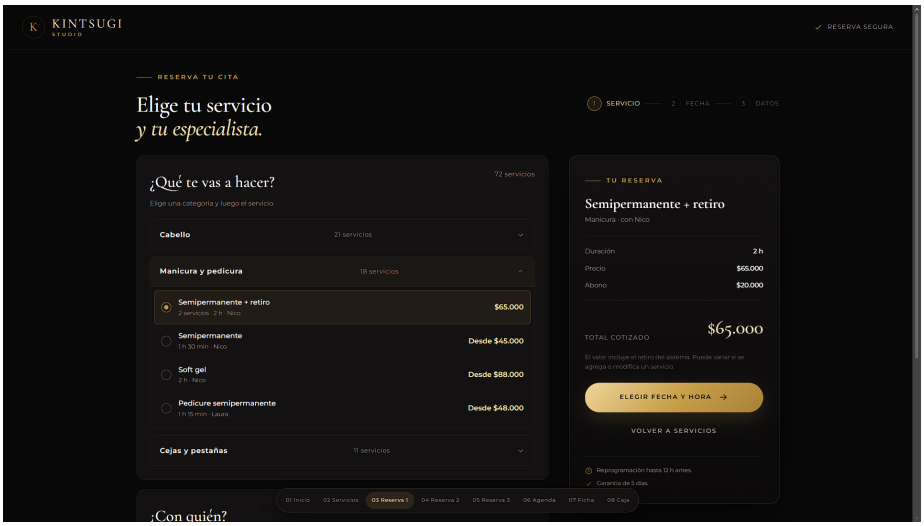
7.2 Services page

Purpose: help the visitor choose a service with confidence. **Elements:** category filter, service cards with name, description, duration, price, and professional; detail panel; booking call to action.



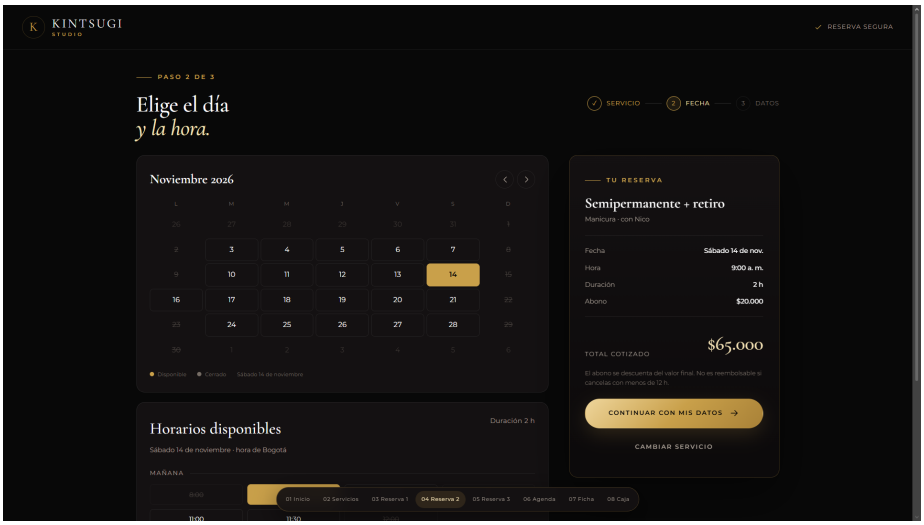
7.3 Booking — step 1: service and professional

Purpose: select what the client wants and who will deliver it. **Elements:** service selector, professional selector with "any available" option, price and duration summary, continue button.



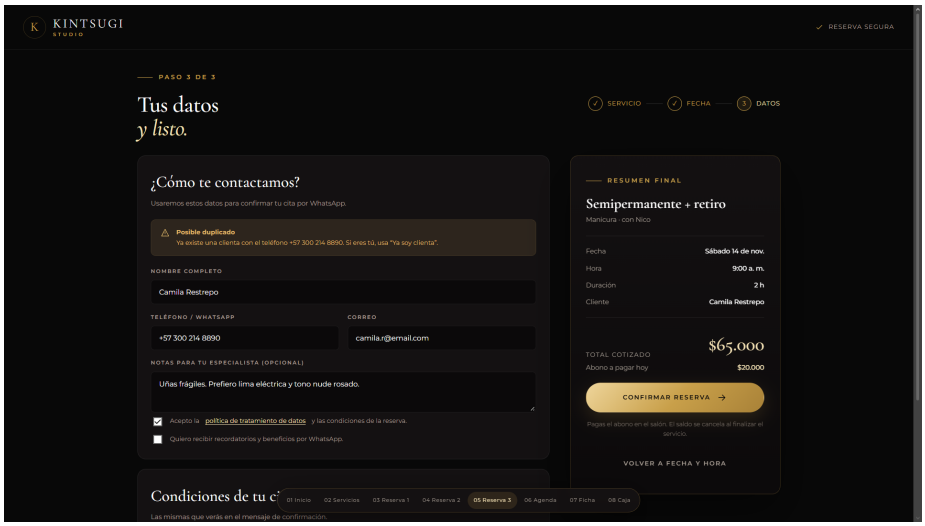
7.4 Booking — step 2: date and time

Purpose: show real availability and capture the slot. **Elements:** calendar, available slots, time zone, temporary hold notice, back and continue buttons.



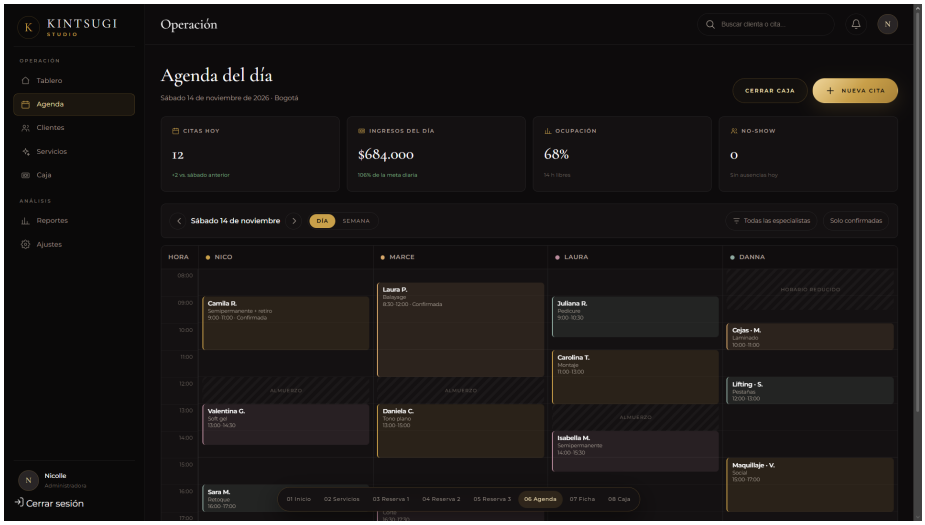
7.5 Booking — step 3: confirmation

Purpose: collect client data and confirm the appointment. **Elements:** name, phone, email, optional notes, marketing consent checkbox, summary, confirm button, policy link.



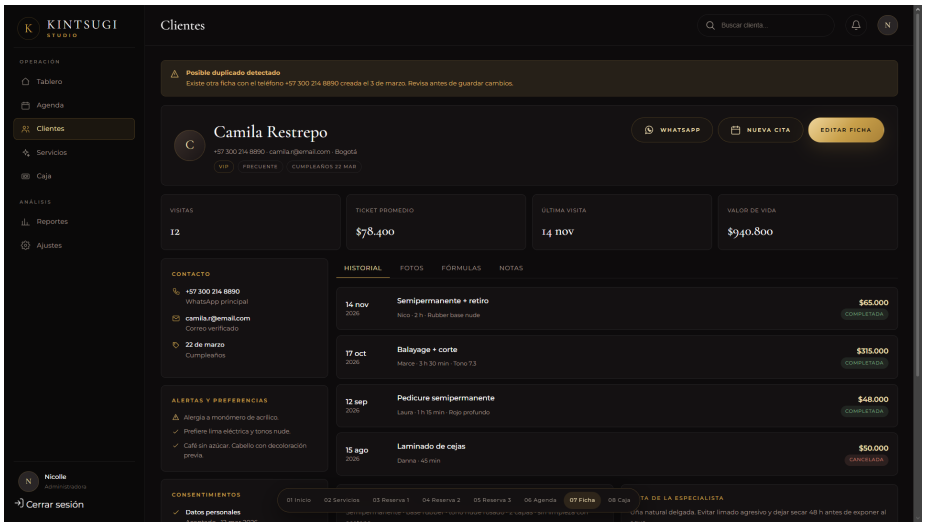
7.6 Administrative agenda

Purpose: give the administrator a reliable view of the day and week. **Elements:** date selector, professional filter, day/week toggle, appointment cards with status, create and move actions, no-show and cancel actions.



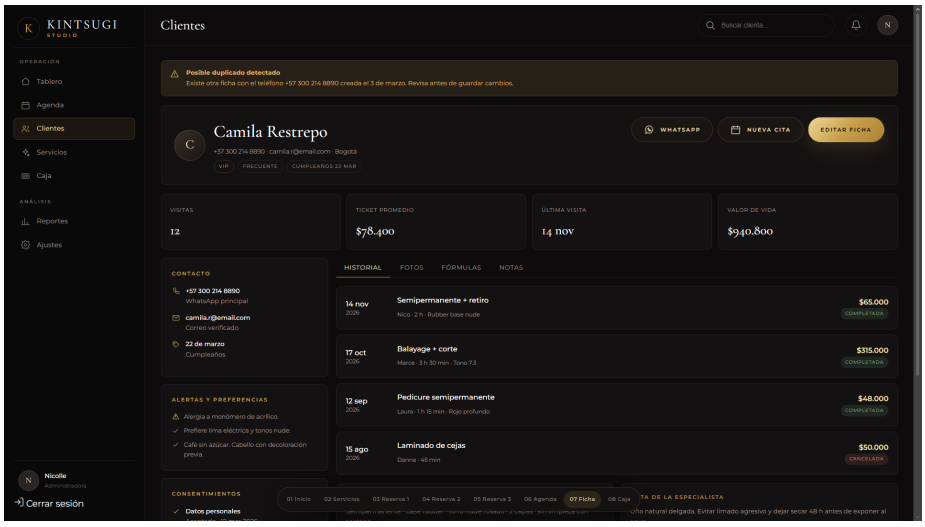
7.7 Client record

Purpose: centralize the client's history and service notes. **Elements:** contact data, appointment history, service items, photographs, formulas, notes, allergies, consent status, duplicate warning.



7.8 Cash closing

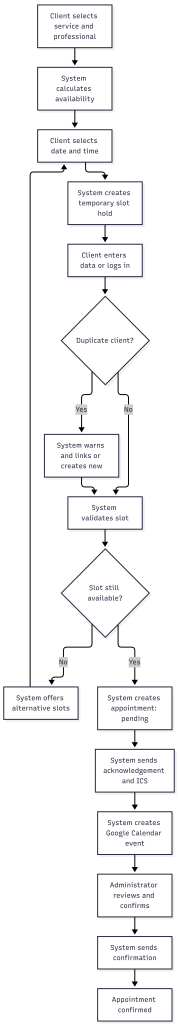
Purpose: close the day and compare the system total with the counted cash. **Elements:** date, income by payment method, expenses by category, counted cash, difference, justification, close button, printable summary.



8. Business Process Models (BPMN)

8.1 Booking process

Participants: Client, System, Administrator, Google Calendar, WhatsApp/Email.

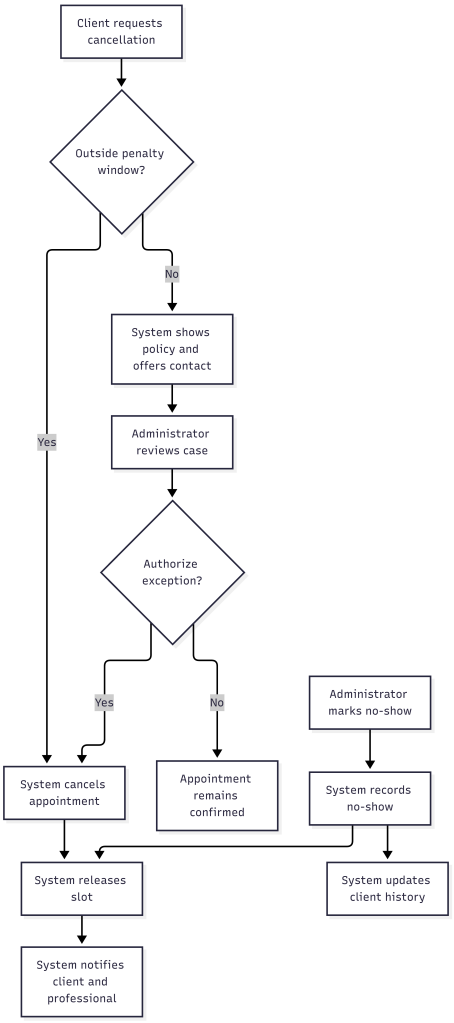


Description: the process starts when the client selects a service and a professional. The system calculates availability and holds the slot while the client completes the form. Before creating the appointment, the system checks for duplicate clients and revalidates the slot inside a transaction. If the slot is no longer available, the client receives alternatives. If the appointment is created, the system sends an acknowledgement, generates the ICS file, and creates the calendar event. The administrator confirms the appointment according to the

studio's policy.

8.2 Cancellation and no-show process

Participants: Client, System, Administrator.

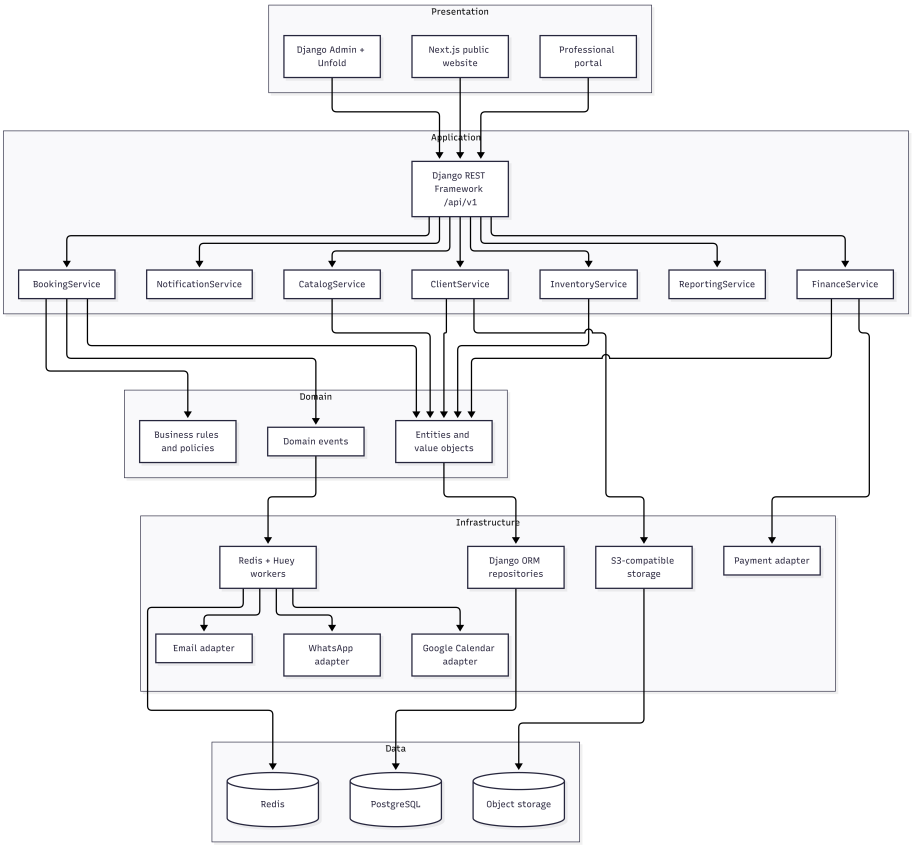


Description: the client can cancel within the policy window. The system cancels the appointment, releases the slot, and notifies the client and the professional. Outside the window, the case goes to the administrator. A no-show is registered by the administrator and affects the client's history and the studio's indicators.

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9. Architecture Diagram

Style: modular monolith with layered architecture and ports/adapters at the integration edges.



Component responsibilities

Component	Responsibility
Next.js public website	Marketing content, service catalog, booking flow, client account.
Professional portal	Own agenda, service history, and reminders.
Django Admin + Unfold	Internal CRM/ERP: agenda, clients, finance, inventory, providers.
Django REST Framework	REST API, validation, authentication, and authorization.
BookingService	Availability, slot hold, appointment creation, rescheduling, cancellation.

Component	Responsibility
CatalogService	Services, categories, prices, and professional assignments.
ClientService	Client records, history, duplicate detection, consent, segmentation.
FinanceService	Cash closing, expenses, commissions, accounts payable, reports.
InventoryService	Items, stock movements, provider invoices.
NotificationService	Email and WhatsApp templates, reminders, birthday messages.
ReportingService	Daily dashboard and monthly indicators.
Redis + Huey	Background jobs: reminders, calendar sync, weekly commissions.
PostgreSQL	Relational source of truth.
Object storage	Client photos, formulas, and documents.

Communication flows

- The browser communicates with the API over HTTPS/JSON.
- The API authenticates with JWT and enforces role and object-level permissions.
- Background jobs run asynchronously and retry failed integrations.
- External integrations are wrapped in adapters so the domain does not depend on vendor SDKs.

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10. Design Decisions

Decision	Rationale
Modular monolith instead of microservices	200–230 clients per month and a two-person team do not justify distributed complexity.
Django + DRF for the backend	Mature ORM, authentication, permissions, admin, and testing; strong fit for OOP and design patterns.
Next.js for the public frontend	Server-side rendering improves SEO and first load for a marketing site.

Decision	Rationale
Django Admin + Unfold for the back-office	Delivers CRM/ERP screens quickly without building dozens of custom CRUD views.
PostgreSQL	Relational integrity for appointments, payments, commissions, and inventory.
Redis + Huey for jobs	Reminders and integrations need scheduling without the operational weight of Celery.
S3-compatible storage	Client photos must survive deployments and be access-controlled.
Adapters for external services	Payment, calendar, and WhatsApp vendors can change without affecting the domain.
Audit log and 2FA	Required by the client for administrators and operators.
Phased roadmap	The full CRM/ERP is a multi-month product; the MVP must solve the operational core first.

Patterns applied

- **Strategy:** commission calculation (55 % / 60 %) and availability policies.
- **Factory:** creation of notifications by channel.
- **Observer:** appointment events trigger calendar sync, reminders, and audit.
- **State:** appointment lifecycle.
- **Facade:** `BookingService` as the single entry point for booking operations.
- **Adapter:** payment gateway, Google Calendar, and WhatsApp integrations.